

A2 (Conditional Independence) — Vulnerability Analysis and Fix

Trigger: Hostile review pointed out that A2 is “almost certainly false, the Achilles’ heel of the entire framework.” **Conclusion:** Diagnosis correct, but fixable. Replace with weak-correlation concentration inequality, degrade the decay rate.

1. Problem Diagnosis

Theorem 1’s proof chain:

A2 (conditional independence) $\rightarrow$ e_m independent $\rightarrow$ Hoeffding concentration $\rightarrow \exp(-2$

If expert errors have conditional correlation $\rho > 0$ (given state s): - CNN: all experts err together on blurry images - MLIP: all experts err together on transition-state configurations - Effective independent expert count: $M_{\text{eff}} \approx M/(1 + (M - 1)\rho)$

At $\rho = 0.3$, $M = 12$ degrades to $M_{\text{eff}} = 2.8$ — the exponential gain nearly vanishes.

CIFAR-10 experimental evidence: Theoretical lower bound F1 0.976 (assumed independence), measured F1=0.617. The gap cannot be explained by μ_s estimation error alone.

AlN experimental evidence: Global bound F1 0.87 matches empirical value. But this only holds when $\rho \lesssim 0.02$ — an extremely weak assumption that requires verification.